P/E Varies Across Countries

Cross-Country Differences in Valuation Multiples

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Outline

Motivation

The Question

Price-earnings ratios vary dramatically across countries.

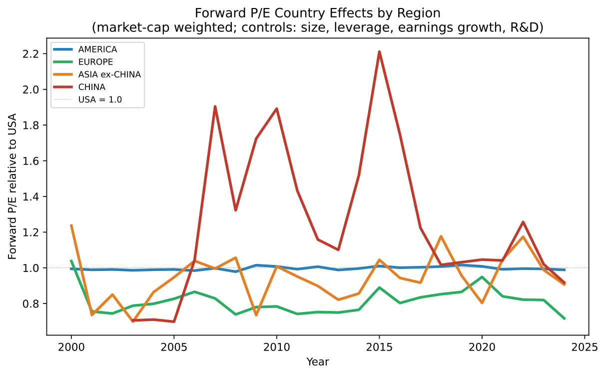
How much of this is firm composition vs. a pure country effect?

Gordon model: $P/E = \dfrac{1}{r - g}$

- Forward P/E uses analyst EPS forecasts $\Rightarrow$ expected growth is controlled
- Residual country effects must reflect **discount rates** or **model misspecification**
- We also measure expected returns directly (analyst price targets)

Approach: We use 200K firm-years across 100+ countries to decompose cross-country P/E differences into discount rate and growth components.

The Puzzle in One Picture



Market-cap-weighted regional aggregates. Year-by-year regressions controlling for firm size, leverage, R&D, growth, and 2-digit SIC industry. USA = 1.0.

Europe has traded at a 10–30% P/E discount to the US for 25 years.

Data

Coverage: 200K Firm-Years, 100+ Countries

	Forward P/E	Exp. Return	LTG
Firm-years	200K	193K	56K
Countries	102	103	68
Years	2000–2025	2000–2025	2000–2025
Firms/year	6K–9K	6K–9K	2K–3K

Key variables:

Forward P/E	IBES price / FY1 EPS consensus
Expected return	Price target / price − 1
MTG	$(\mathrm{EPS}_{\mathrm{FY2}} - \mathrm{EPS}_{\mathrm{FY1}}) / \lvert\overline{\mathrm{EPS}}\rvert$

Top 10: USA 84K, JPN 31K, GBR 14K, CAN 9K, AUS 8K, IND 7K, CHN 4K, FRA 4K, DEU 4K, KOR 4K

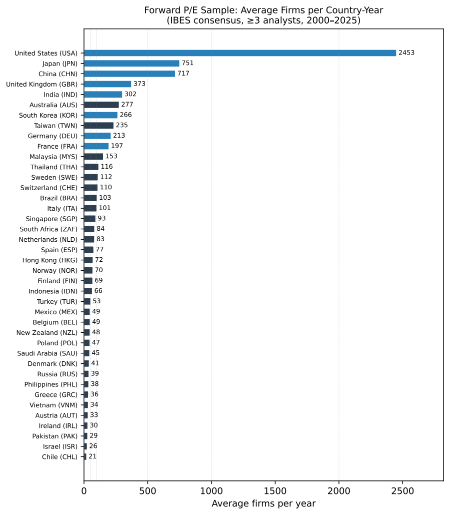
Key advantage: no market cap required.

Forward P/E = IBES price / IBES EPS forecast, both in same IBES currency ⇒ no FX or market cap needed.

Non-US coverage from 2000 (60–80 countries), not just from 2007.

P/E winsorized at 1/99%. Expected return at 2.5/97.5%.
All regressions use log(P/E).
≥3 analysts required for consensus.
FE: Country×Year + SIC2×Year throughout.

Sample Size by Country



Country P/E Effects Are Large

Firm-Level Panel: Forward P/E with Progressive Controls

Dep. var: $\log(\text{Forward P/E FY1})_{i,t}$. Country×Year + SIC2×Year FE. SE clustered by country.

	(A) Base	(B) + R&D	(C) + MTG	(C′) + LTG
Leverage	−0.12*	−0.10*	−0.18*	−0.32***
log(Assets)	−0.04***	−0.03**	−0.01	−0.03***
Lag ΔE/|E|	−0.003**	−0.002*	+0.001	−0.005***
R&D / Assets		+2.14***	+1.86***	+1.88***
MTG (FY2−FY1)			+1.17***	
IBES LTG (%)				+0.011***
N	197K	197K	189K	52K
R^2	0.28	0.29	**0.53**	0.37
Countries	102	102	101	70
$\sigma(\hat{\gamma}_c)$	**0.36**	**0.36**	**0.34**	**0.35**

Key: MTG doubles R^2 (0.29 → 0.53) but $\sigma(\hat{\gamma}_c)$ barely moves (0.36 → 0.34). Growth expectations explain *within*-country P/E variation, not *between*-country differences. Lag $\Delta E/|E| = (E_t - E_{t-1})/|E_{t-1}|$.

What Does $\sigma(\hat{\gamma}_c) = 0.34$ Mean?

In economic terms:

$\sigma = 0.34$ in $\log(P/E) \approx$ **40% P/E premium** for a one-σ country.

Concrete examples (vs. USA):

Country	P/E†	ER‡
Korea	0.61×	+33pp
UK	0.79×	+2pp
France	0.81×	≈ 0
Germany	0.85×	+1pp
India	0.90×	+12pp
Japan	1.03×	+11pp
China	1.05×	+6pp

†$\exp(\hat{\gamma}_c)$, USA= 1. ‡Analyst expected return vs USA (PTG/P−1).

Comparison: Country vs. Industry

FE Structure	R^2
SIC2 (Industry)	0.08
Country	0.09
SIC2×Year	0.15
Country×Year	0.17
Both (baseline)	**0.28**
+ Firm controls	**0.53**

Country effects are **as large as industry effects**.

Country×Year FE add **12pp** beyond SIC2×Year, of which 5pp is time-varying.

Country P/E effects are as large as industry effects and firm controls barely shrink them.

Can discount rates or growth expectations explain this?

Decomposition: Discount Rate vs. Growth

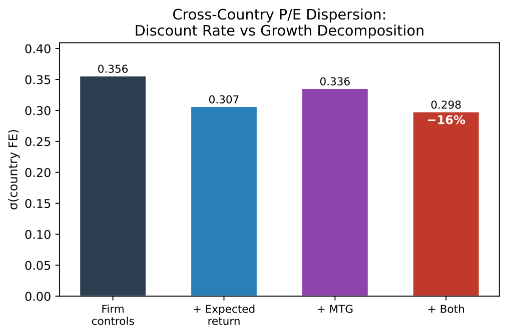
Expected Return Regressions: The Discount-Rate Side

Dep. var: Expected return = Price target/Price − 1. Country×Year + SIC2×Year FE. SE clustered by country.

	(A) Base + MTG	**(B)** + R&D	**(C′)** + LTG
Leverage	+0.05*	+0.05*	+0.10***
log(Assets)	−0.02***	−0.02***	−0.01***
Lag $\Delta E/\lvert E\rvert$	−0.003***	−0.003***	−0.006***
MTG (FY2−FY1)	−0.03**	−0.03**	
R&D / Assets		+0.07	+0.44***
IBES LTG (%)			+0.002***
N	162K	162K	51K
R^2	0.29	0.29	0.35
Countries	100	100	68
$\sigma(\hat{\gamma}_c)$	**0.16**	**0.16**	**0.09**

Signs consistent with risk pricing: leverage ↑ ER, size ↓ ER. Country ER effects ($\sigma = 0.16$) are large but **much smaller** than P/E effects ($\sigma = 0.34$). Korea +33pp, Japan +11pp, India +12pp vs USA.

How Much Do Expected Returns and Growth Explain?



Gordon model: $P/E = 1/(r-g)$

$$\log(P/E) \approx -r + g$$

Expected return (PTG/P−1) proxies for r. MTG proxies for g.

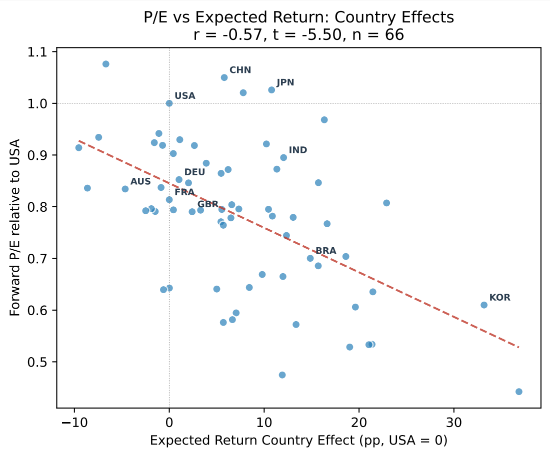
Decomposition of $\sigma(\hat{\gamma}_c)$:

- Expected return absorbs **14%**
- MTG absorbs **6%**
- Both absorb **16%** (overlap: not additive)

⇒ 84% of P/E dispersion unexplained.

exp_ret: -0.61 ($t = -20.8$); MTG: $+1.16$ ($t = +46.9$)

P/E vs. Expected Return Across Countries



Time-averaged country FE (USA $= 0$ for ER, $= 1$ for P/E).

$r = -0.57, \quad t = -5.50, \quad n = 66$

Strong negative correlation — the right direction and **highly significant**.

Notable:

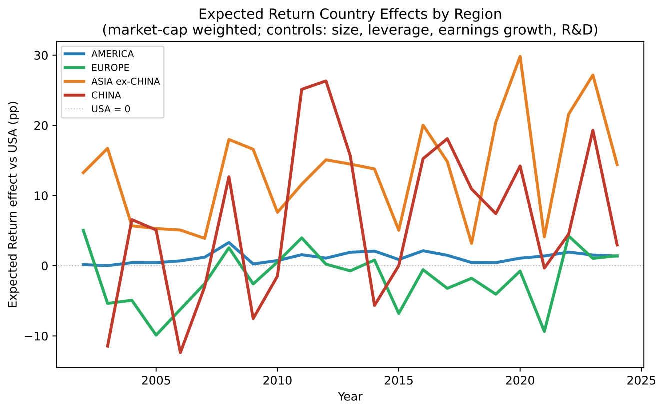
- Korea: highest ER, lowest P/E
- Japan, China: high P/E, moderate ER
- France, Germany, UK: low P/E, near-zero ER

Yet ER absorbs only 14% of $\sigma(\hat{\gamma}_c)$.

Why? Analyst bias may understate true ER differences. Or the gap is not about discount rates at all.

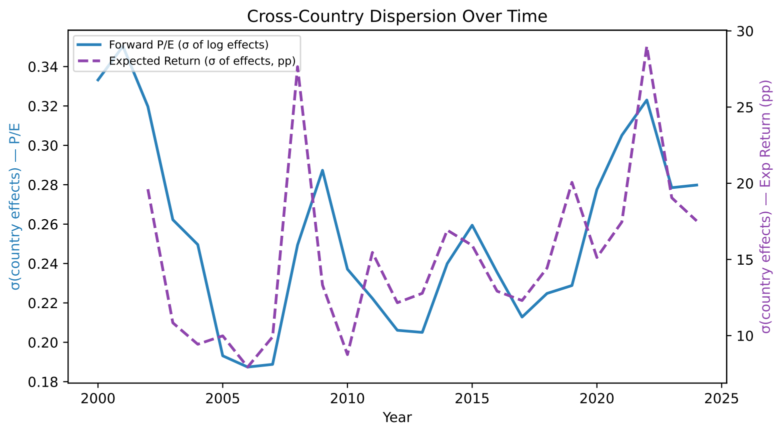
Persistence and Cross-Country Predictors

Expected Return: Regional Effects Over Time



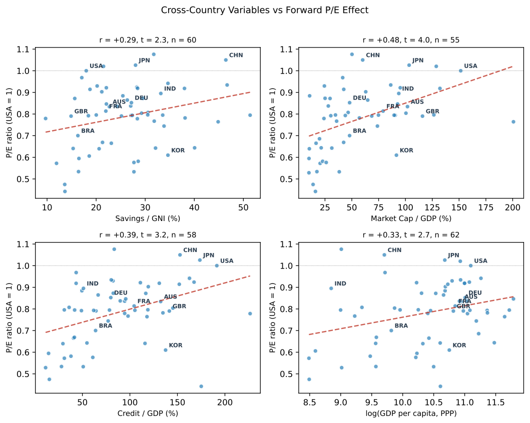
Market-cap-weighted regional aggregates. Year-by-year regressions (firm controls + 2-digit SIC industry FE). USA = 0. Asia ex-China: persistently +10–30pp above USA. Europe: near zero. America (ex-USA): close to USA.

Cross-Country Dispersion Over Time



σ of country effects across countries in each year. Dispersion in both P/E and expected return is substantial and persistent — not converging.

What Predicts the Country P/E Effect?

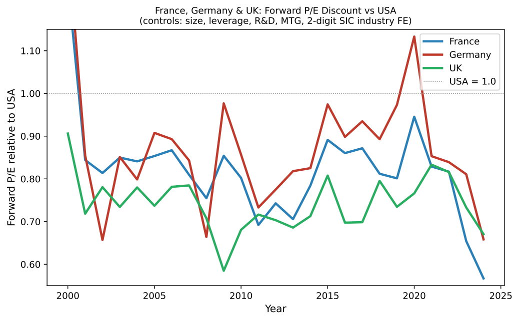


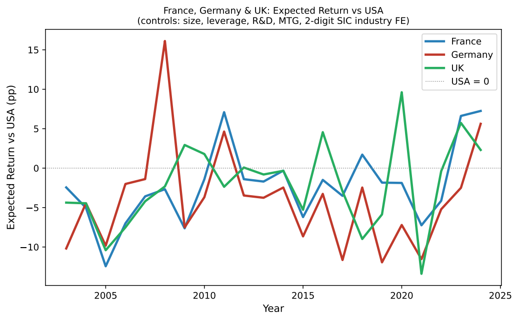
Dep. var: country FE $\hat{\gamma}_c$ (time-averaged, USA = 0, 66 ctry)

	P/E		Exp. Ret.	
Predictor	r	t	r	t
Mkt Cap / GDP	**+.48**	**3.96**	−.22	−1.63
Credit / GDP	**+.39**	**3.20**	−.15	−1.10
log(GDP/cap)	**+.33**	**2.69**	**−.31**	**−2.57**
Savings / GNI	**+.29**	**2.28**	−.15	−1.13
Anti-Self-Dealing	+.24	1.69	−.18	−1.28

Financial development and income predict P/E. ER signs are consistent but mostly insignificant — only log(GDP/cap) significant.

France, Germany & UK: P/E Discount and Expected Return





Year-by-year regressions (size, leverage, R&D, MTG, SIC2 industry FE). P/E discount widens from parity to 0.57–0.67× (2024), yet ER gap remains small (+2–7pp). **Discount rates do not explain the European P/E discount.**

Summary

Three Facts About Cross-Country P/E

1. **Country effects are as large as industry effects.**

   After controlling for firm size, leverage, R&D, and growth with Country$\times$Year + SIC2$\times$Year FE: $\sigma(\hat{\gamma}_c) = 0.34$ in $\log(P/E)$, equivalent to a **40% P/E premium** per standard deviation. Europe trades at 0.7–0.9$\times$ USA for 25 years. The gap is **widening**.

2. **Discount rates go in the right direction but explain little.**

   P/E and expected return FE correlate strongly ($r = -0.57$), but ER absorbs only **14%** of $\sigma(\hat{\gamma}_c)$. Growth expectations absorb 6%. Together, 16%.

3. **Financial development predicts the gap. Governance does not.**

   Mkt cap/GDP ($r = 0.48$), credit/GDP ($r = 0.39$), and log GDP/cap ($r = 0.33$) predict higher P/E. Anti-self-dealing is null ($t = 1.7$), contrary to La Porta et al. (2002).

84% of cross-country P/E variation remains unexplained.

Research Questions

1. **What is the missing factor?**
   Persistent country effects unexplained by firm characteristics, growth, or risk. Behavioral? Cultural? Tax? Liquidity?
2. **Do analyst expected returns predict realized returns?**
   High analyst ER $\Rightarrow$ high realized returns $=$ risk-based. Otherwise $=$ behavioral.
3. **Does the "Korea discount" reflect governance?**
   Korea: P/E 0.61$\times$, ER $+$33pp. Classical governance story, but anti-self-dealing index is null ($t = 1.7$).

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994) — country dominates returns. We extend to *valuation levels*.
- Griffin–Karolyi (1998) — industry effects vary across sectors.

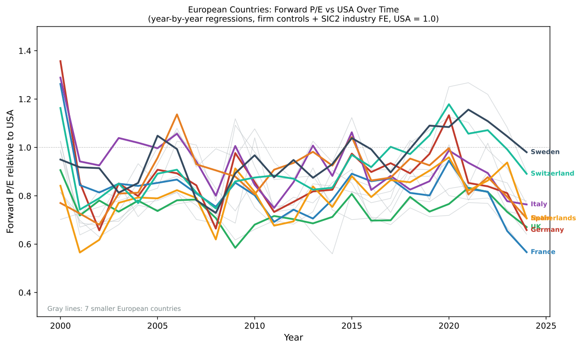
Institutions and cost of capital:

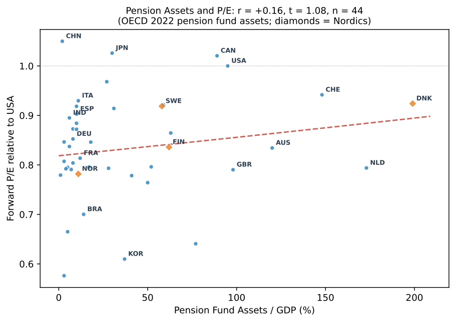
- La Porta et al. (2002) — investor protection raises Tobin's Q. We find: **null for forward P/E**.
- Djankov et al. (2008) — anti-self-dealing index. **Null** as P/E predictor ($t = 1.7$).
- Hail–Leuz (2006) — institutions reduce cost of equity. Our expected return evidence is mixed.
- Stulz (2005) — "twin agency problems" prevent equalization.

Analyst forecasts and pricing:

- Da–Warachka (2011) — analyst forecasts and the cross-section of expected returns.
- Bradshaw et al. (2012) — analysts' forecasts reflect risk differentials.

P/E Dispersion Within Europe





Within Europe: Denmark and Sweden recently at or above USA; France, Germany, UK at 0.6–0.8×. **Funded pension hypothesis:** countries with large pension fund assets (Nordics, Switzerland) have higher P/E — but correlation is weak globally ($r = 0.16$, $t = 1.08$). Netherlands (173% pension/GDP, 0.79× P/E) is a counterexample.

Appendix

Data Sources and Construction

Source	Obs	Description
Compustat Global + NA	1.2M	Annual fundamentals: assets, debt, earnings, R&D, SIC industry
IBES Summary (FY1/FY2)	3.1M	Median analyst EPS forecast ($\geq$3 analysts)
IBES Panel Summary	5.4M	Prices contemporaneous with consensus
IBES Price Targets	3.2M	Median analyst 12-month price target
IBES Long-Term Growth	507K	Consensus long-term EPS growth forecast (%)
OECD STES + BIS	1.6K	10Y govt bond yield (47 ctry) + policy rate gap-fill (10 ctry)
World Bank	5.7K	CPI inflation rate; 241 countries, 2000–2025
IMF WEO	3.4K	5-year GDP growth forecast; 196 countries, 2008–2026
World Bank (cross-sec.)	varies	Savings/GNI, mkt cap/GDP, credit/GDP, GDP/cap PPP

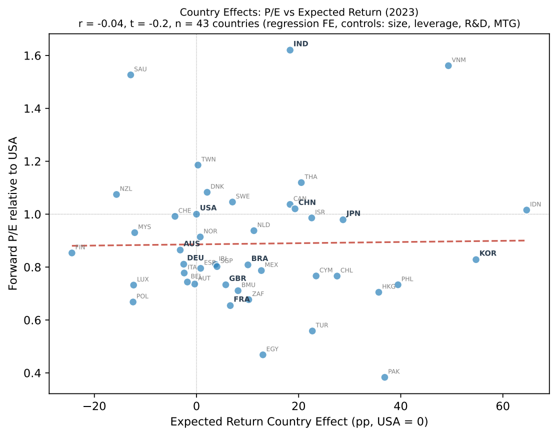
Key measurement choices:

- Forward P/E = IBES price / IBES EPS forecast (same currency $\Rightarrow$ **currency-invariant**)
- Expected return = Price target / Price − 1 (from `ptgsum` / `pansum`)
- Consensus and price contemporaneous at `statpers`; latest before fiscal year end

Variable Definitions

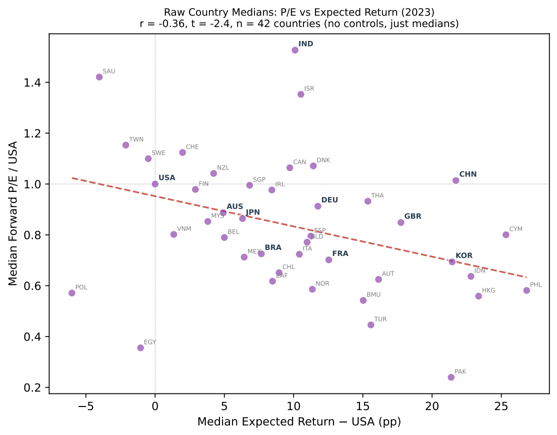
Variable	Construction	Coverage	Source
Firm-level (panel)			
Forward P/E	IBES price / FY1 EPS consensus	200K, 102 ctry	IBES
Expected return	Price target / price − 1	193K, 103 ctry	IBES
LTG	Consensus long-term EPS growth (%)	56K, 68 ctry	IBES
Leverage	(Long-term + short-term debt) / assets	197K	Compustat
R&D / Assets	R&D expense / total assets (0 if missing)	197K	Compustat
SIC2 Industry	2-digit SIC (e.g. 20=Food, 35=Machinery, 73=Business Services)	88%, 74 industries	Compustat
Country × year (macro panel)			
Bond yield 10y	10Y govt bond (OECD, 47 ctry) + BIS policy rate (10 ctry)	57 ctry	OECD + BIS
Inflation	CPI inflation rate, annual % (YoY)	241 ctry	World Bank
IMF growth 5y	Avg 5yr-ahead real GDP growth forecast	196 ctry	WEO vintages
Cross-sectional (time-averaged)			
Savings / GNI	Gross national savings / GNI (2005–23)	214 ctry	World Bank
Mkt Cap / GDP	Stock market capitalization / GDP (2010–23)	129 ctry	World Bank
Credit / GDP	Private credit / GDP (2010–23)	232 ctry	World Bank
Anti-Self-Dealing	Index (0–1), anti-self-dealing protection	46 ctry	Djankov (2008)

2023 Snapshot: Regression-Based Country Effects



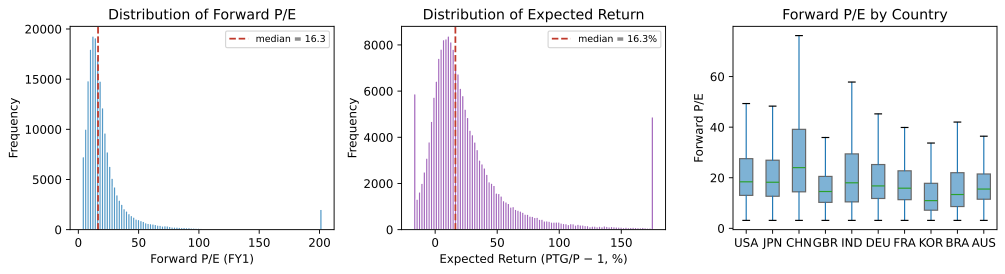
Year-by-year regression FE (firm controls + 2-digit SIC industry FE). USA $= (0, 1)$. Korea: low P/E, high ER. Japan: high P/E. European markets cluster below USA.

2023 Snapshot: Raw Country Medians (No Controls)



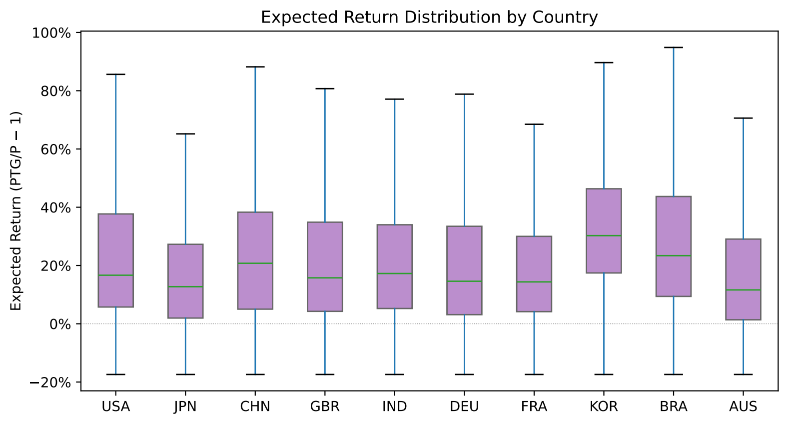
Raw median P/E and expected return by country (no firm-level controls). Similar pattern to regression FE, confirming that country effects are not driven by model specification.

Forward P/E and Expected Return: Distributions



Left: Forward P/E distribution (winsorized 1/99%, log-transformed in regressions). **Center:** Expected return (PTG/P−1, winsorized 2.5/97.5%). Median = 16% — analysts are systematically optimistic. **Right:** Forward P/E by country (box plots, no fliers).

Expected Return Distribution by Country



Korea median ≈ 30%, Japan ≈ 13%, USA ≈ 17%. Analyst optimism is pervasive but cross-country differences in expected returns are real and persistent.

IBES Linking and Coverage Details

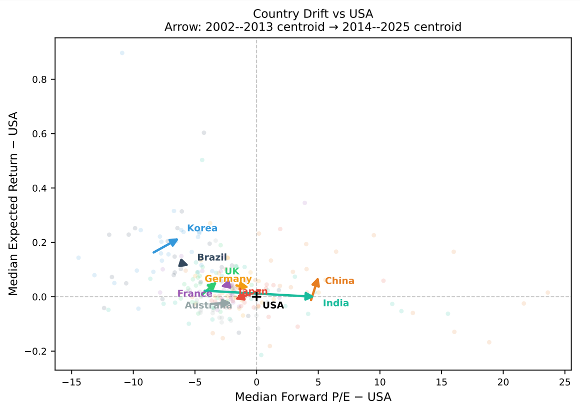
Linking: US via ICLINK + CCM; International via `ibtic` (41K pairs). Total: 62K tickers.

Price data: IBES `pansum` table. Price and consensus contemporaneous at `statpers` (3rd Thursday).

Forward P/E = `pansum.price` / `statsum.fy1_eps` (both in IBES currency $\Rightarrow$ no FX noise). UK pence: ratio is correct.

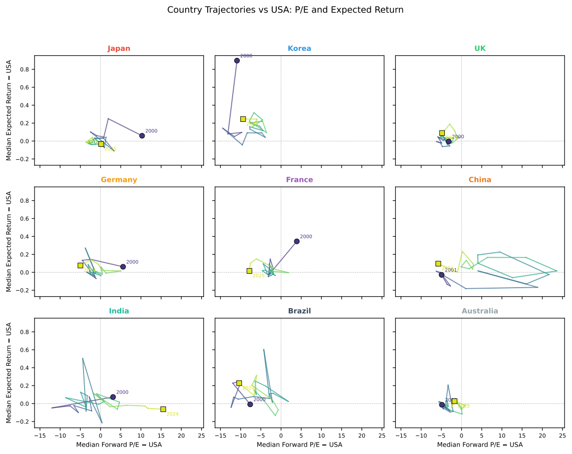
Outliers: Countries $<$100 obs excluded from 2nd stage. Year-by-year: $\geq$20 firms/country.

Country Trajectories in (P/E, Expected Return) Space



Median P/E and expected return minus USA median, per country-year ($\geq$20 firms). Arrow: 2002–2013 $\rightarrow$ 2014–2025 centroid. Korea: low P/E, high ER vs USA. India drifts to high P/E.

Country Trajectories: Individual Paths



Country-year medians minus USA. Circle = 2000, Square = 2025. Color: early (dark) to late (light).